

Block Chain

What is a Consensus Algorithm?

A **consensus algorithm** is the set of rules that allows thousands of blockchain nodes to **agree on a single, valid version of the blockchain** without trusting one another.

Imagine there are **10,000 computers** in a blockchain network.

If Alice sends 2 BTC to Bob:

- How do all computers know this transaction is genuine?
- How do they agree that Alice actually owns the BTC?
- How do they ensure the same transaction isn't accepted twice?

The answer is **consensus**.

Without consensus:

- Different nodes could have different versions of the blockchain.
 - Double spending would be possible.
 - The blockchain would fail.
-

Why Do We Need Consensus?

Imagine a class of 100 students.

The teacher asks:

┆ "Who scored the highest marks?"

Everyone gives different answers.

There must be **one agreed answer**.

Similarly,

Thousands of blockchain nodes need **one agreed blockchain**.

Consensus ensures:

- Everyone has the same ledger.
 - Fake transactions are rejected.
 - Double spending is prevented.
 - No central authority is required.
-

What Does Your Website Need?

Your website is titled

Consensus Algorithms in Blockchain — A Comparative Explorer

It is **not** simply about listing algorithms.

Instead,

it is a **comparison platform**.

Users should be able to choose an algorithm like

Proof of Work

and instantly see

- How it works
- Security
- Scalability
- Decentralization
- Coins using it
- Programming language
- Layer
- Compatibility with other blockchains

all updated together.

Section 1Algorithm Overview

This explains

What is this algorithm?

Example:

Proof of Work

Definition

Miners compete to solve a difficult mathematical puzzle.

The first miner to solve it

wins the right to create the next block.

Reward:

- New coins
 - Transaction fees
-

What should be shown?

Algorithm Name

Example

Proof of Work (PoW)

Core Mechanism

Miners solve cryptographic puzzles.

Plain Language Explanation

Instead of saying

"hash-based computational puzzle"

say

Imagine 100 students solving a very difficult math puzzle.

The first student to solve it

gets to write the next page of the class notebook.

That notebook is the blockchain.

Step-by-step

Example

Transaction Created

↓

Miners Collect Transactions

↓

Solve Puzzle

↓

Winner Found

↓

Block Created

↓

Network Verifies

↓

Blockchain Updated

Every algorithm will have

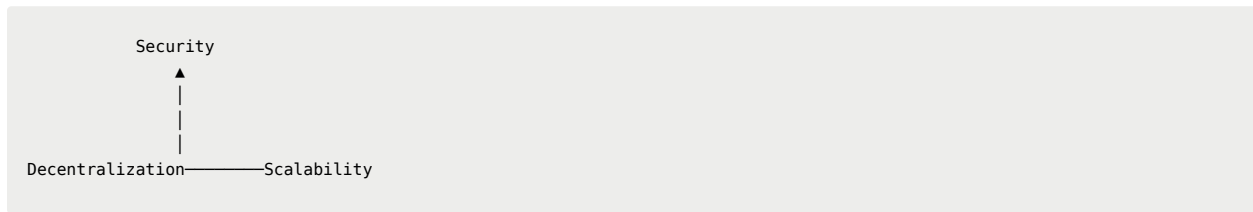
its own steps.

Section 2Blockchain Trilemma

One of the most important concepts.

It was proposed by Vitalik Buterin.

A blockchain cannot maximize all three simultaneously.



Every blockchain compromises somewhere.

Pillar 1

Scalability

Meaning

How many transactions can the blockchain process?

Measured using

TPS

Transactions Per Second

Examples

Bitcoin

7 TPS

Ethereum

15–30 TPS (Layer 1)

Solana

Thousands of TPS under favorable conditions

Also measure

Block Creation Time

Bitcoin

10 minutes

Ethereum

~12 seconds

Solana

Sub-second target block production

Questions to compare

How fast?

How many TPS?

How much congestion?

Pillar 2

Security

How difficult is it to attack?

Examples

51% Attack

One entity controls most validating power.

Sybil Attack

One attacker creates many fake identities.

Questions

Can someone cheat?

Can someone rewrite history?

How expensive is an attack?

Pillar 3

Decentralization

Who controls the blockchain?

Questions

How many validators?

Can anyone join?

Is participation permissionless?

Bitcoin

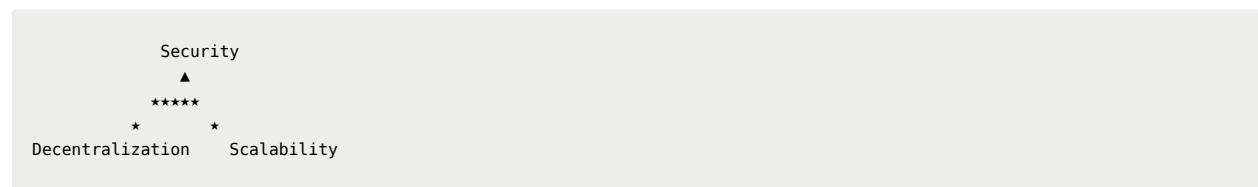
Very decentralized.

Private blockchain

Not decentralized.

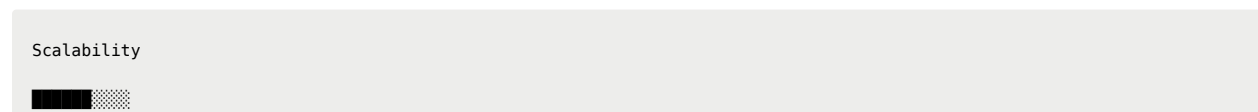
How to display?

Radar Chart



or

Scorecards



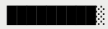
6/10

Security



10/10

Decentralization



9/10

Section 3Real-world Blockchain Mapping

Users should know

Which blockchain uses which algorithm?

Example

Blockchain	Consensus
Bitcoin	Proof of Work
Ethereum	Proof of Stake
Solana	Proof of History + PoS
Cardano	Ouroboros PoS
Avalanche	Avalanche Consensus
Algorand	Pure PoS

Also explain

Why?

Example

Bitcoin

Uses PoW

Reason

Maximum security.

Ethereum

Uses PoS

Reason

Reduce energy usage.

Solana

Uses PoH

Reason

Extremely high speed.

Section 4 Smart Contract Languages

Each blockchain supports different programming languages.

Examples

Ethereum

Solidity

Solana

Rust

Move-based chains

Move

Cardano

Plutus (Haskell-based)

BNB Chain

Solidity

Avalanche

Solidity (through the EVM-compatible C-Chain)

Show

Ethereum

↓

PoS

↓

Solidity

Users understand

Consensus

↓

Blockchain

↓

Programming Language

Section 5 Layer Classification

Many students confuse Layer 1 and Layer 2.

Layer 1

The main blockchain.

Examples

Bitcoin

Ethereum

Solana

Cardano

Avalanche

Layer 1 handles

Consensus

Security

Block production

Native token

Layer 2

Built on top of Layer 1.

Purpose

Increase speed.

Reduce fees.

Examples

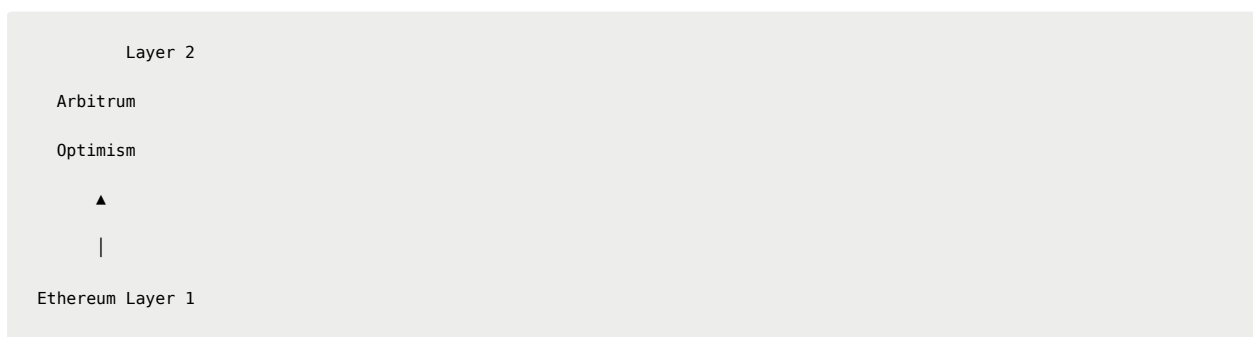
Arbitrum

Optimism

Base

Polygon PoS is often described as a sidechain rather than a traditional Layer 2, so explain its architecture carefully if you include it.

Diagram



Section 6Compatibility Matrix

Probably the most interactive part.

Suppose

Bitcoin

Ethereum

Solana

Cardano

Make

	BTC	ETH	SOL	ADA
BTC	✓	✗	✗	✗
ETH	✗	✓	✗	✓
SOL	✗	✗	✓	✗
ADA	✗	✓	✗	✓

Meaning

Do they use similar consensus?

Ethereum

PoS

Cardano

PoS

Compatible

Bitcoin

PoW

Ethereum

PoS

Different

Not compatible.

Heatmap

Green

Same consensus

Red

Different consensus

Section 7Filtering

This is what makes the website interactive.

Example

Click

Proof of Stake

Everything changes.

Overview

↓

PoS

Radar

↓

PoS scores

Coins

↓

Ethereum

Cardano

Polkadot

Language

↓

Solidity

Plutus

Rust (depending on chain)

Layer

↓

Layer 1

Compatibility

↓

Updated matrix

One click updates the entire page.

What Algorithms Should You Cover?

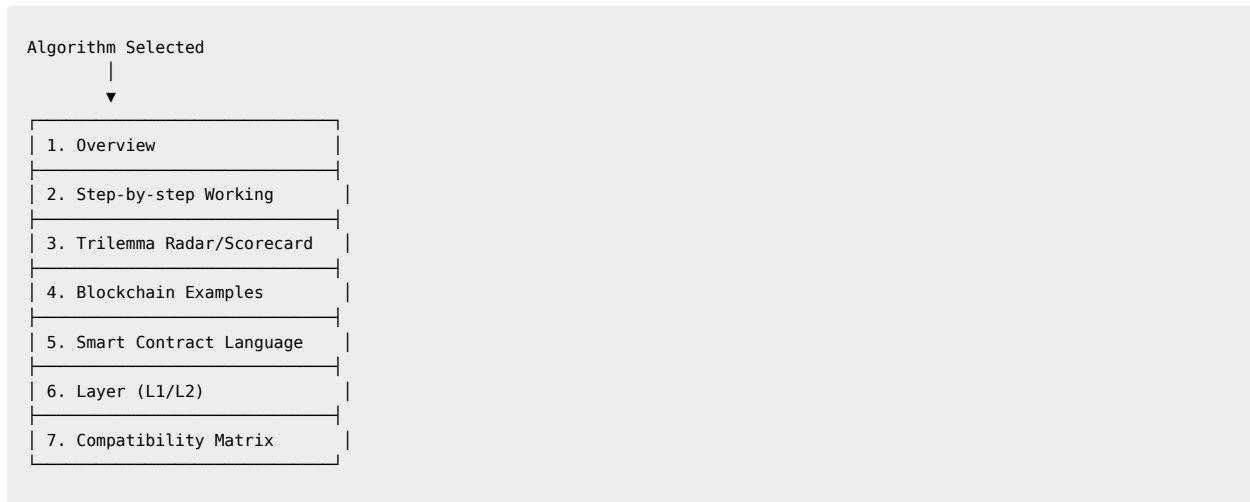
For a comprehensive explorer, I recommend these **10 major consensus algorithms**:

Algorithm	Representative Blockchain
Proof of Work (PoW)	Bitcoin

Algorithm	Representative Blockchain
Proof of Stake (PoS)	Ethereum
Delegated Proof of Stake (DPoS)	EOS
Proof of Authority (PoA)	Private/EVM networks
Proof of History (PoH + PoS)	Solana
Practical Byzantine Fault Tolerance (PBFT)	Hyperledger Fabric (enterprise platform)
Proof of Elapsed Time (PoET)	Enterprise networks
Proof of Capacity (PoC)	Burst
Avalanche Consensus	Avalanche
Ouroboros (PoS variant)	Cardano

Final Website Structure

When you build the website, each algorithm should drive **all** of these synchronized panels:



This structure ensures the site is educational, visually engaging, and satisfies all the requirements of your assignment.

The "**How difficult is it to attack?**" metric measures **how resistant a blockchain's consensus algorithm is to malicious attacks**. It evaluates how easy or difficult it is for an attacker to manipulate the blockchain, steal funds, or disrupt normal operations.

The two most common attacks are **51% Attack** and **Sybil Attack**.

1. 51% Attack Definition

A **51% attack** occurs when a single individual, organization, or group gains control of **more than 50% of the network's validating power**.

The "validating power" depends on the consensus algorithm:

- **Proof of Work (PoW):** More than 50% of the mining (hashing) power.

- **Proof of Stake (PoS):** More than 50% of the total staked cryptocurrency.

With majority control, the attacker can influence which transactions are accepted.

How it Works

Suppose a blockchain has 100 miners.

Miner	Mining Power
Honest miners	49%
Attacker	51%

The attacker now controls the majority of the network.

They can:

1. Create their own blocks.
 2. Make their blockchain version grow faster.
 3. Force the network to accept their version of the blockchain.
-

Example

Alice has **10 BTC**.

1. Alice sends **10 BTC** to Bob.
2. Bob delivers the product after seeing the transaction confirmed.
3. Meanwhile, Alice secretly mines another version of the blockchain where that payment never happened.
4. Since Alice controls **51% of the mining power**, her private chain becomes longer.
5. The network accepts Alice's longer chain.

Result:

- Bob loses his money.
- Alice keeps both the bitcoins and the product.

This is called **double spending**.

What an Attacker Can Do

- ✓ Reverse their own transactions.
 - ✓ Perform double spending.
 - ✓ Prevent some transactions from being confirmed.
 - ✓ Delay block creation.
-

What an Attacker Cannot Do

- ✗ Create new coins.
- ✗ Steal coins from someone else's wallet.
- ✗ Change old cryptographic signatures.

Difficulty

- **Small blockchain:** Easy
 - **Medium blockchain:** Difficult
 - **Large blockchain (e.g., Bitcoin):** Extremely difficult because acquiring over 50% of the global hashing power would require enormous hardware, electricity, and cost.
-

2. Sybil AttackDefinition

A **Sybil attack** occurs when an attacker creates **many fake identities (nodes)** to gain excessive influence over the network.

Instead of increasing mining power, the attacker increases the **number of fake participants**.

How it Works

Imagine a blockchain network with 10 nodes.

```
Node A
Node B
Node C
Node D
Node E
Node F
Node G
Node H
Node I
Node J
```

An attacker creates **100 fake nodes**.

```
Fake Node 1
Fake Node 2
Fake Node 3
...
Fake Node 100
```

Now the attacker appears to be many different participants.

If the system simply counts the number of nodes, the attacker could dominate voting or communication.

Example

Suppose a blockchain votes on whether to accept a block.

Normal users:

```
10 Honest Votes
```

Attacker:

The attacker wins the vote even though there is only one real attacker.

What a Sybil Attacker Can Do

- Disrupt consensus.
- Spread false information.
- Delay transactions.
- Isolate honest nodes (Eclipse attack).
- Influence voting systems.

Difficulty

It depends on the consensus mechanism:

- **Proof of Work (PoW):** Very difficult, because each identity must also provide computational power.
- **Proof of Stake (PoS):** Difficult, because each identity must lock up cryptocurrency as stake.
- **Permissionless peer-to-peer networks without strong identity checks:** Easier, since creating many fake identities can be inexpensive.

Comparison

Feature	51% Attack	Sybil Attack
Main Goal	Control the blockchain	Control the network using many fake identities
Requires	Majority of mining/staking power	Large number of fake nodes
Cost	Very expensive	Usually cheaper
Primary Target	Consensus mechanism	Network participation and voting
Can Cause Double Spending	✅ Yes	❌ Usually No
Common in	PoW and PoS blockchains	Peer-to-peer networks and weakly protected blockchain networks

Simple Real-World Analogy

51% Attack

Imagine a classroom with **100 students** voting.

If one group controls **51 students**, they can decide the outcome of every vote.

Sybil Attack

Imagine one person creates **100 fake student IDs** and uses them to vote multiple times.

Although only one real person exists, they appear to be many different voters.

Summary

- **51% Attack:** An attacker gains **majority computational or staking power**, allowing them to rewrite recent blockchain history, reverse their own transactions, and perform **double spending**.
- **Sybil Attack:** An attacker creates **many fake identities or nodes** to manipulate communication, voting, or network behavior. Modern blockchain consensus mechanisms make Sybil attacks much harder by tying participation to costly resources such as computing power (PoW) or cryptocurrency stake (PoS).

1. Proof of Work (PoW)

1. OverviewDefinition

Proof of Work (PoW) is the **first and most widely known blockchain consensus algorithm**, introduced by Satoshi Nakamoto in 2008 for Bitcoin.

It allows thousands of computers (miners) to agree on which transactions are valid by requiring them to solve a difficult mathematical puzzle.

The first miner to solve the puzzle earns the right to add the next block to the blockchain and receives a reward.

Why was PoW created?

Before Bitcoin, digital money had a major problem called **double spending**.

Example:

Alice owns only **1 BTC**.

She tries to send that same Bitcoin simultaneously to Bob and Charlie.

Without a trusted central authority, the network needs a way to determine **which transaction is valid**.

Proof of Work solves this by making it computationally expensive to create blocks and by allowing the network to agree on one valid chain.

Core Principle

"The computer that proves it has performed sufficient computational work earns the right to create the next block."

The "work" consists of solving a cryptographic puzzle.

2. Core Mechanism

PoW relies on three major concepts.

A. Cryptographic Hash Function

Every block is hashed using functions such as **SHA-256** (Bitcoin).

Example

```
Input:
Hello

Hash:
185f8db32271fe25...
```

Even a one-letter change produces a completely different hash.

B. Nonce

A **nonce** is a number miners continuously change while searching for a valid block hash.

Example

```
Nonce = 101

Hash

89ABCD...

❌ Invalid
```

```
Nonce = 102

Hash

000000F91A...

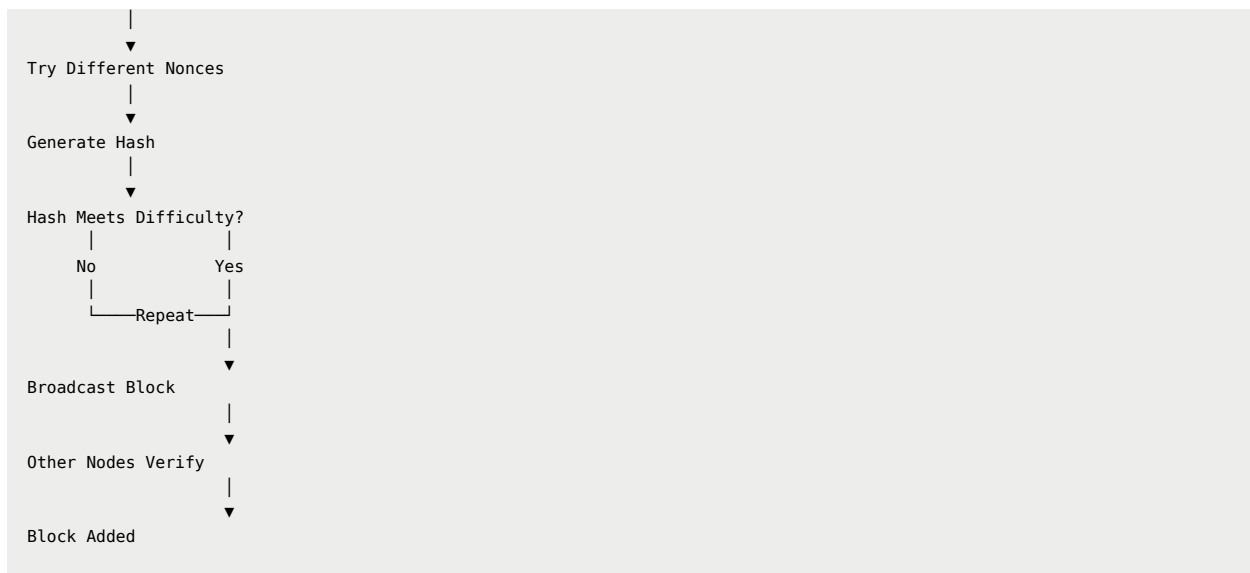
✅ Valid
```

C. Mining

Mining is the process of repeatedly trying different nonce values until the generated hash satisfies the network difficulty target.

3. Step-by-Step Working

```
User Creates Transaction
    |
    ▼
Transaction Broadcast
    |
    ▼
Miners Collect Transactions
    |
    ▼
Create Candidate Block
```



Example

Imagine 5 miners competing.

Miner A

Hash

1AFD...



Miner B

Hash

4C91...



Miner C

Hash

00000009AF



Winner

Miner C receives

- Block reward
- Transaction fees

4. Participants

PoW consists of

Users

Create transactions.

Miners

Collect transactions.

Solve cryptographic puzzles.

Produce new blocks.

Full Nodes

Verify blocks.

Maintain blockchain.

Reject invalid blocks.

5. Selection Process

Who creates the next block?

Answer

The miner that first finds a valid hash satisfying the current network difficulty.

Selection depends entirely on computational work.

6. Reward Mechanism

Successful miners receive

Block Reward

Newly minted cryptocurrency.

Example

Bitcoin block reward (varies over time due to halvings).

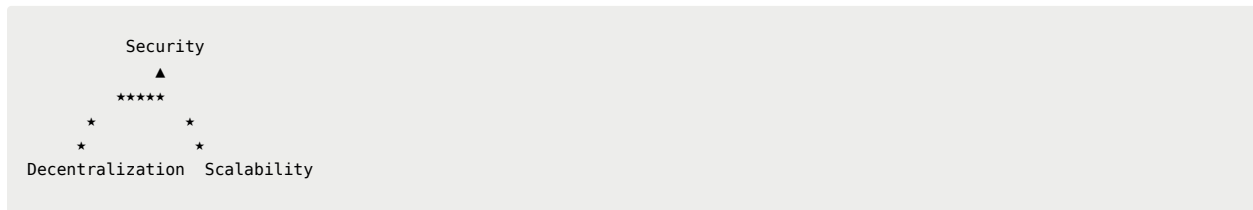
Transaction Fees

Fees paid by users whose transactions are included.

7. Blockchain Trilemma Analysis

Property	Score	Reason
Scalability	3/10	Low TPS and longer block intervals
Security	10/10	Extremely secure and battle-tested
Decentralization	9/10	Anyone with suitable hardware can participate

Radar



8. Performance

Parameter	Value
TPS	~7 (Bitcoin)
Block Time	~10 minutes
Finality	Probabilistic (often ~6 confirmations for high assurance)
Energy Usage	Very High
Latency	High

9. Attack Resistance

51% Attack
If someone controls more than 50% of the network's mining power, they may be able to reorganize recent blocks and perform attacks such as double spending.

Difficulty

Very expensive on large networks like Bitcoin.

Sybil Attack

Creating many fake identities alone is not enough.

The attacker also needs significant mining power.

Therefore

PoW is highly resistant.

Double Spending

Very difficult once a transaction has sufficient confirmations.

10. Advantages

- Highest level of security.
- Fully decentralized.
- Proven over many years.
- Simple validation rules.

11. Disadvantages

- Very high electricity consumption.
- Specialized mining hardware can become expensive.
- Low TPS.
- Longer confirmation times.

12. Real-world Blockchain Mapping

Blockchain	Cryptocurrency	Why PoW?
Bitcoin	BTC	Maximum security and decentralization
Litecoin	LTC	Faster block times than Bitcoin while retaining PoW
Dogecoin	DOGE	Community-driven currency using PoW
Monero	XMR	Privacy-focused PoW network

13. Smart Contract Languages

Blockchain	Smart Contract Language
Bitcoin	Bitcoin Script
Litecoin	Bitcoin Script
Dogecoin	Not designed for advanced smart contracts
Monero	No general-purpose smart contracts

14. Layer Classification

Blockchain	Layer
Bitcoin	Layer 1
Litecoin	Layer 1
Dogecoin	Layer 1
Monero	Layer 1

PoW primarily secures **Layer 1** blockchains.

15. Environmental Impact

Metric	Rating
Electricity	★★★★★
Carbon Footprint	High
Sustainability	Low

16. Best Use Cases

- Digital currency
- Long-term value storage
- High-security payment networks

17. Summary

Property	Value
Consensus	Proof of Work
Mining	Yes
Validators	Miners
Energy	Very High
Speed	Low
Security	Excellent
Decentralization	High

2. Proof of Stake (PoS)

1. OverviewDefinition

Proof of Stake (PoS) is a consensus algorithm in which validators are selected to create and validate new blocks based primarily on the amount of cryptocurrency they have **staked** (locked as collateral), rather than on computational power.

Instead of solving puzzles, validators demonstrate commitment to the network by locking their coins.

Why was PoS created?

PoW provides strong security but consumes significant energy.

PoS was designed to:

- Reduce electricity consumption.

- Improve transaction throughput.
 - Lower hardware requirements.
 - Enable broader participation in validation.
-

Core Principle

"The more stake you commit to the network (along with protocol-specific selection rules), the greater your chance of being selected to validate a block."

Unlike PoW,

No mining occurs.

Validators replace miners.

2. Core Mechanism

PoS uses

A. Staking

Validators lock cryptocurrency as collateral.

Example

Validator A

Locks

32 ETH

B. Validator Selection

The protocol selects a validator according to its rules.

The exact method differs across blockchains and may consider stake, randomness, validator performance, and other factors.

C. Validation

The selected validator proposes the next block.

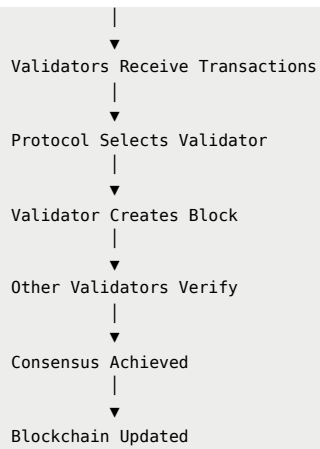
Other validators verify it.

3. Step-by-Step Working

User Creates Transaction



Transaction Broadcast



Example

Validator A

Stake

500 Coins

Validator B

Stake

10,000 Coins

Validator C

Stake

25,000 Coins

Selected

4. Participants

Users
Create transactions.

Validators

Stake coins.

Validate blocks.

Earn rewards.

Full Nodes

Verify blockchain data.

5. Selection Process

The protocol chooses a validator based on its staking mechanism.

Unlike PoW, validators do not compete by solving computational puzzles.

6. Reward Mechanism

Validators receive

- Staking rewards.
- Transaction fees.

If validators violate protocol rules, some PoS systems impose **slashing**, reducing a portion of the validator's staked funds.

7. Blockchain Trilemma Analysis

Property	Score	Reason
Scalability	8/10	Faster block production and higher throughput than PoW
Security	9/10	Strong security when the staking mechanism is well designed
Decentralization	8/10	Generally high, though very large stakeholders may have greater influence

8. Performance

Parameter	Value
TPS	Typically higher than PoW (varies by blockchain)
Block Time	Usually a few seconds to tens of seconds
Finality	Often faster than PoW (depends on protocol)
Energy Usage	Very Low

9. Attack Resistance51% Attack

An attacker would generally need control of a very large portion of the staked cryptocurrency, making attacks economically costly.

Sybil Attack

Simply creating many identities does not help because validator influence depends on stake rather than identity count.

Nothing-at-Stake

Some early PoS designs faced this concern, but modern PoS protocols include mechanisms such as slashing and finalized checkpoints to mitigate it.

10. Advantages

- Very energy efficient.
 - Higher TPS than PoW.
 - Faster confirmations.
 - No specialized mining hardware required.
-

11. Disadvantages

- Wealth concentration can influence validator selection.
 - Staked funds may be locked for a period.
 - Protocol design is more complex than PoW.
-

12. Real-world Blockchain Mapping

Blockchain	Cryptocurrency	Why PoS?
Ethereum	ETH	Reduced energy use and scalable validation
Cardano	ADA	Energy-efficient research-driven PoS
Polkadot	DOT	Shared security across parachains
Tezos	XTZ	On-chain governance with staking

13. Smart Contract Languages

Blockchain	Language
Ethereum	Solidity, Vyper
Cardano	Plutus (Haskell-based)
Polkadot	Rust (Substrate ecosystem)
Tezos	Michelson, SmartPy, LIGO

14. Layer Classification

All the above examples are **Layer 1** blockchains.

Many **Layer 2** networks (such as rollups built on Ethereum) inherit security from a PoS Layer 1 rather than running their own independent PoS consensus.

15. Environmental Impact

Metric	Rating
Electricity	★
Carbon Footprint	Very Low
Sustainability	Excellent

16. Best Use Cases

- Smart contract platforms
 - Decentralized finance (DeFi)
 - NFTs
 - High-throughput blockchain ecosystems
-

17. Summary

Property	Value
Consensus	Proof of Stake
Mining	No
Validators	Yes
Energy	Very Low
Speed	High
Security	Very High
Decentralization	High (depends on validator distribution)

3. Delegated Proof of Stake (DPoS)

1. OverviewDefinition

Delegated Proof of Stake (DPoS) is an advanced version of **Proof of Stake (PoS)** where token holders **vote to elect a small number of trusted delegates (also called witnesses or block producers)** who are responsible for validating transactions and producing new blocks.

Instead of every validator participating directly, the community elects representatives to perform validation on their behalf.

It was introduced by **Daniel Larimer** in 2014 to improve blockchain speed and scalability.

Why was DPoS created?

Traditional PoS improves energy efficiency but can become slow when thousands of validators participate.

DPoS was created to:

- Increase transaction speed
 - Reduce block creation time
 - Improve scalability
 - Introduce democratic voting for validator selection
-

Core Principle

| "Token holders vote for delegates, and the elected delegates create and validate blocks."

Unlike PoS:

- Everyone does **not** validate blocks.
 - Only elected delegates perform validation.
-

2. Core Mechanism

DPoS has three important components:

A. Token Holders

Every coin holder has voting power.

Generally, more tokens mean greater voting influence, depending on the protocol.

B. Delegates (Witnesses)

Delegates are elected by token holders.

Their responsibilities:

- Validate transactions
 - Produce blocks
 - Maintain the network
-

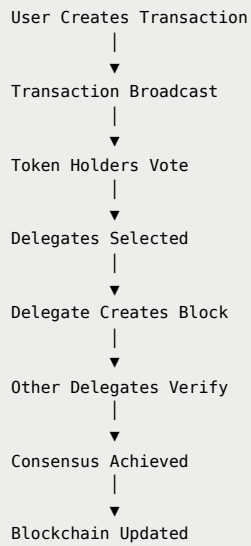
C. Voting

If delegates perform poorly or act maliciously, users can vote them out.

New delegates are elected.

This makes DPoS a **representative democracy**.

3. Step-by-Step Working



Example

Imagine

10,000 users

↓

Vote

↓

21 delegates elected

↓

Those 21 delegates continuously produce blocks.

If Delegate 5 becomes dishonest,

↓

Users vote again.

↓

Delegate 5 is replaced.

4. Participants

Create transactions.

Token Holders

Vote for delegates.

Delegates (Witnesses)

Validate transactions.

Produce blocks.

Maintain the blockchain.

Full Nodes

Store blockchain.

Verify blocks.

5. Selection Process

Selection occurs through voting.

Example

```
10000 Users
↓
Voting
↓
Delegate A
4500 votes
Delegate B
4200 votes
Delegate C
4100 votes
Top delegates selected
```

Unlike PoW,

No mining competition.

Unlike PoS,

Not every validator participates.

6. Reward Mechanism

Delegates receive

- Block rewards
- Transaction fees

Some delegates share rewards with voters.

7. Blockchain Trilemma Analysis

Property	Score	Reason
Scalability	9/10	Small validator set enables very fast block production.
Security	8/10	Secure when honest delegates are elected, but collusion is a concern.
Decentralization	6/10	Only a limited number of delegates create blocks.

8. Performance

Parameter	Value
TPS	Hundreds to thousands (depends on blockchain)
Block Time	Usually 0.5–3 seconds
Finality	Fast
Energy Usage	Very Low

9. Attack Resistance51% Attack

Harder because delegates are elected,
but collusion among delegates is possible.

Sybil Attack

Voting reduces the impact of fake identities.

Simply creating many accounts does not provide additional influence without voting power.

Delegate Collusion

Several delegates could cooperate to censor transactions.

This is the major criticism of DPoS.

10. Advantages

- Extremely fast
 - Low transaction fees
 - Energy efficient
 - Easy governance
 - High scalability
-

11. Disadvantages

- Less decentralized
 - Voting participation may be low
 - Delegate collusion is possible
 - Wealthy token holders may have greater influence
-

12. Real-world Blockchain Mapping

Blockchain	Cryptocurrency	Why DPoS?
EOS	EOS	High throughput using elected block producers
TRON	TRX	Fast transactions through Super Representatives
Steem	STEEM	Community governance with witnesses
Lisk	LSK	JavaScript-based blockchain with delegated validation

13. Smart Contract Languages

Blockchain	Language
EOS	C++
TRON	Solidity
Lisk	JavaScript / TypeScript

14. Layer Classification

All major DPoS blockchains operate as

✓ Layer 1

because they maintain their own consensus and blockchain.

15. Environmental Impact

Metric	Rating
Electricity	★
Carbon Footprint	Very Low
Sustainability	Excellent

16. Best Use Cases

- High-speed payments
 - Gaming blockchains
 - Social media platforms
 - Enterprise applications
 - High-TPS decentralized applications
-

17. Summary

Property	Value
Consensus	Delegated Proof of Stake
Mining	No
Validators	Elected Delegates
Voting	Yes
Energy	Very Low
Speed	Very High
Security	High
Decentralization	Medium

4. Proof of Authority (PoA)

1. OverviewDefinition

Proof of Authority (PoA) is a consensus algorithm where **pre-approved, trusted validators (authority nodes)** are responsible for validating transactions and creating new blocks.

Instead of relying on computational work or staked cryptocurrency, PoA relies on the **identity and reputation** of validators.

Only approved authorities can participate.

Why was PoA created?

PoA was designed for:

- Private blockchains
- Consortium blockchains
- Enterprise applications

where participants are already known and trusted.

It prioritizes speed and efficiency over open participation.

Core Principle

| "Trusted identities validate the blockchain."

The network trusts approved validators rather than anonymous miners or stakers.

2. Core Mechanism

PoA relies on:

A. Authority Nodes

Only approved organizations or individuals can validate blocks.

B. Identity Verification

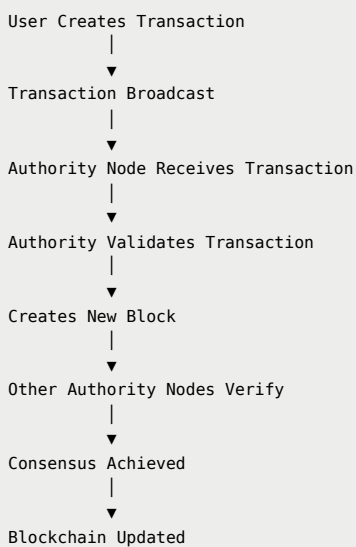
Validators are selected based on verified identities and reputation.

C. Reputation

If an authority behaves maliciously,
it can be removed from the network.

Its reputation is at risk.

3. Step-by-Step Working



Example

Imagine

10 banks

Each bank operates one authority node.

Only those banks can create blocks.

No outsider can become a validator.

4. Participants

Users
Create transactions.

Authority Nodes

Validate transactions.

Produce blocks.

Maintain blockchain.

Organizations

Approve and manage authority nodes.

5. Selection Process

Validators are selected

through identity verification and approval,

not through mining or staking.

Example

Company Approves

↓

Validator A

Validator B

Validator C

↓

Only these validators create blocks.

6. Reward Mechanism

In many enterprise PoA networks:

- No cryptocurrency reward.

- Validators participate because they are part of the organization or consortium.

Public PoA networks may include transaction fee incentives.

7. Blockchain Trilemma Analysis

Property	Score	Reason
Scalability	10/10	Very few validators enable extremely fast block creation.
Security	8/10	Strong when validators are trusted, but depends on validator honesty.
Decentralization	3/10	Validation is limited to a small approved group.

8. Performance

Parameter	Value
TPS	Hundreds to thousands
Block Time	Often 1–5 seconds
Finality	Fast
Energy Usage	Extremely Low

9. Attack Resistance51% Attack

Unlikely in permissioned networks,

but compromise of a majority of authority nodes could threaten the system.

Sybil Attack

Very resistant.

Unknown identities cannot simply join the validator set.

Insider Attack

The biggest risk.

If trusted validators become malicious,
they can damage the network.

10. Advantages

- Extremely fast
- Minimal energy usage
- High throughput
- Simple consensus

- Ideal for permissioned environments

11. Disadvantages

- Low decentralization
 - Requires trust in validators
 - Less suitable for open public cryptocurrencies
 - Possibility of censorship by authority nodes
-

12. Real-world Blockchain Mapping

Blockchain	Cryptocurrency	Why PoA?
VeChain	VET	Enterprise supply-chain network using authority validators
XinFin XDC Network	XDC	Hybrid enterprise-focused blockchain
POA Network	POA	Built specifically around the PoA model

Private enterprise networks based on platforms like Hyperledger Besu can also use PoA.

13. Smart Contract Languages

Blockchain	Language
VeChain	Solidity
XDC Network	Solidity
Hyperledger Besu (PoA)	Solidity (EVM-compatible)

14. Layer Classification

Most PoA deployments are

✔ Layer 1 permissioned blockchains.

Some organizations also use PoA for private sidechains.

15. Environmental Impact

Metric	Rating
Electricity	★
Carbon Footprint	Very Low
Sustainability	Excellent

16. Best Use Cases

- Supply chain management
 - Banking consortia
 - Government record systems
 - Healthcare
 - Enterprise blockchain networks
-

17. Summary

Property	Value
Consensus	Proof of Authority
Mining	No
Staking	No
Validators	Approved authority nodes
Energy	Extremely Low
Speed	Very High
Security	High (in trusted environments)
Decentralization	Low

Key Difference: DPoS vs. PoA

Feature	DPoS	PoA
Validator selection	Elected by token holders	Approved by an organization or consortium
Network type	Usually public	Usually private or consortium
Governance	Democratic voting	Centralized approval
Decentralization	Medium	Low
Speed	Very High	Very High
Best suited for	Public high-performance blockchains	Enterprise and permissioned blockchains

5. Proof of History (PoH)

1. OverviewDefinition

Proof of History (PoH) is a cryptographic technique that creates a **verifiable record of time** before transactions are processed. Unlike traditional consensus algorithms that require validators to communicate extensively to agree on the order of transactions, PoH provides a trusted timeline, allowing validators to process transactions more efficiently.

Important Note:

PoH is not a standalone consensus algorithm. It is a **cryptographic clock** that works together with **Proof of Stake (PoS)**. The most well-known implementation is on the Solana blockchain.

Why was PoH created?

Traditional consensus algorithms spend considerable time agreeing on **when** a transaction occurred and **in what order** transactions should be processed.

This communication introduces delays.

PoH was developed to:

- Eliminate the need for validators to constantly synchronize time.
 - Increase transaction throughput.
 - Reduce communication overhead.
 - Improve scalability while maintaining security.
-

Core Principle

|"Instead of asking validators to agree on time, create a cryptographic proof that time has passed."

Think of PoH as a **cryptographic clock** that timestamps every event.

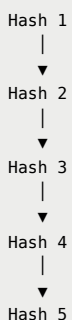
2. Core Mechanism

PoH is based on a **Verifiable Delay Function (VDF)**.

A VDF performs sequential cryptographic computations that cannot be parallelized.

Each output depends on the previous output, creating a chain of cryptographic timestamps.

Example:



Because every hash depends on the previous one, anyone can verify the order in which events occurred.

Components of PoHA. Verifiable Delay Function (VDF)

A function that requires sequential computation.

It proves that a measurable amount of time has passed.

B. Timestamp

Every transaction receives a cryptographic timestamp.

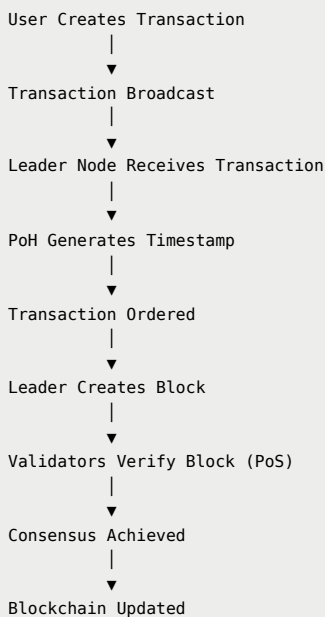
Validators immediately know the correct order of transactions.

C. Proof of Stake

PoH determines **when** events occurred.

PoS determines **who** is allowed to produce the next block.

3. Step-by-Step Working



Example

Imagine three transactions arrive almost simultaneously.

Alice → Bob
Charlie → David
Emma → Frank

Without PoH:

Validators must communicate to determine the order.

With PoH:

Transaction A

↓

Timestamp 105

↓

Transaction B

↓

Timestamp 106

↓

Transaction C

↓

Timestamp 107

The order is already established before validation begins.

4. ParticipantsUsers

Create transactions.

Leader Node

Generates the PoH sequence.

Orders transactions.

Produces blocks.

Validators

Verify:

- Timestamp sequence
 - Transactions
 - Blocks
-

Full Nodes

Maintain blockchain.

Verify historical data.

5. Selection Process

Unlike PoW,
No mining.
Unlike DPoS,
No voting.
Selection follows the **Proof of Stake** mechanism.
Once selected,
the leader generates the PoH sequence and produces blocks.

6. Reward Mechanism

Validators receive:

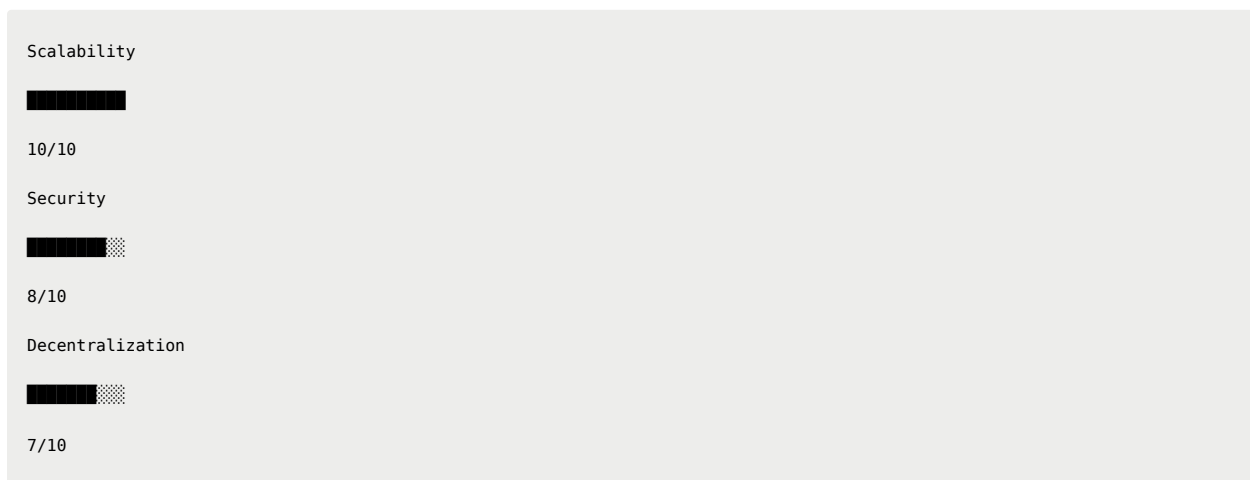
- Staking rewards.
- Transaction fees.

The leader also receives rewards for successful block production.

7. Blockchain Trilemma Analysis

Property	Score	Reason
Scalability	10/10	Extremely high throughput because transaction ordering is predetermined.
Security	8/10	Strong when combined with PoS, though security depends on the overall protocol.
Decentralization	7/10	Public participation is possible, but hardware requirements are relatively high.

Visual Scorecard



8. Performance

Parameter	Value
TPS	Thousands (implementation-dependent)
Block Time	Approximately 400–600 milliseconds (Solana target)
Finality	A few seconds
Energy Usage	Very Low
Latency	Very Low

9. Attack Resistance51% Attack

Because Solana combines PoH with PoS,

an attacker would generally need control over a significant portion of the network's stake rather than simply computational power.

Sybil Attack

Creating fake identities alone is ineffective.

Validators require stake and must follow protocol rules.

Double Spending

PoH's ordered timeline helps reduce conflicting transaction ordering.

Final confirmation still depends on PoS consensus.

Time Manipulation

PoH is specifically designed to make the passage of time verifiable.

Changing timestamps would invalidate the cryptographic sequence.

10. Advantages

- Extremely fast transaction processing.
 - Very high TPS.
 - Low transaction fees.
 - Energy efficient.
 - Efficient transaction ordering.
 - Reduced communication between validators.
 - Excellent scalability.
-

11. Disadvantages

- More complex than traditional consensus algorithms.
- Depends on PoS for validator selection.
- Requires powerful hardware for optimal performance.
- Primarily associated with the Solana ecosystem.

12. Real-world Blockchain Mapping

Blockchain	Cryptocurrency	Why PoH?
Solana	SOL	Achieves very high throughput using PoH for transaction ordering combined with PoS for validation.

At present, PoH is most strongly associated with Solana and is not widely adopted by major independent Layer 1 blockchains.

13. Smart Contract Languages

Blockchain	Language
Solana	Rust
Solana	C
Solana	C++ (supported through the underlying runtime/tooling in some development workflows)

Rust is the primary language used for Solana smart contract (program) development.

14. Layer Classification

Blockchain	Layer
Solana	Layer 1

PoH operates directly within Solana's Layer 1 consensus architecture.

15. Environmental Impact

Metric	Rating
Electricity	★
Carbon Footprint	Very Low
Sustainability	Excellent

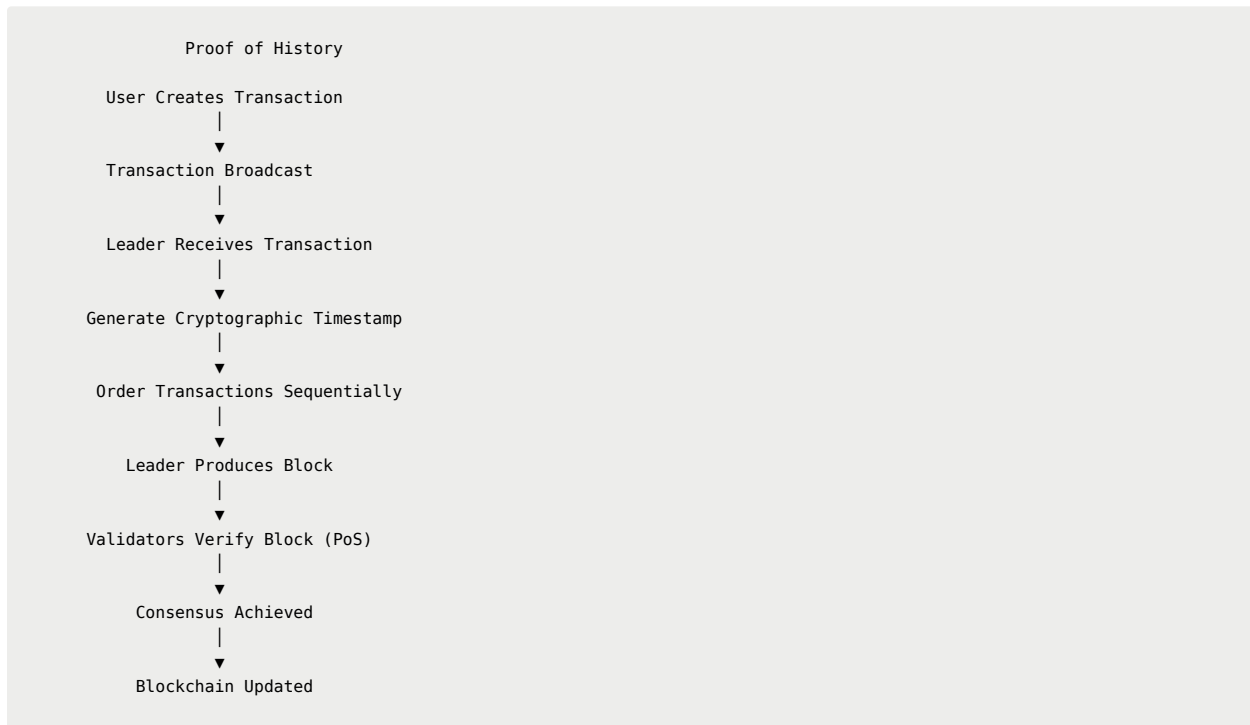
Compared with PoW,

PoH consumes significantly less energy because it does not require competitive mining.

16. Best Use Cases

- High-speed payment systems
- Decentralized Finance (DeFi)
- NFT marketplaces
- Blockchain gaming
- High-frequency trading applications
- Large-scale decentralized applications (dApps)

17. Complete Flow Diagram



18. Summary Table

Property	Value
Consensus	Proof of History + Proof of Stake
Mining	No
Validators	Yes
Time Mechanism	Verifiable Delay Function (VDF)
Transaction Ordering	Automatic cryptographic timestamps
Energy Usage	Very Low
Speed	Extremely High
TPS	Thousands (implementation-dependent)

Property	Value
Block Time	~400–600 ms (Solana target)
Security	High (combined with PoS)
Decentralization	Medium to High
Layer	Layer 1
Primary Blockchain	Solana

19. Key Points for Comparison

Feature	PoW	PoS	DPoS	PoA	PoH
Uses Mining	✓	✗	✗	✗	✗
Uses Staking	✗	✓	✓ (voting-based delegates)	✗	✓ (through PoS)
Uses Voting	✗	✗	✓	✗	✗
Uses Trusted Validators	✗	✗	Elected	✓	✗
Uses Cryptographic Clock	✗	✗	✗	✗	✓
Energy Usage	Very High	Very Low	Very Low	Very Low	Very Low
Speed	Low	High	Very High	Very High	Extremely High
Example Blockchain	Bitcoin	Ethereum	EOS	VeChain	Solana

Simple way to remember:

- **PoW** → *Prove your computing power.*
- **PoS** → *Prove your stake.*
- **DPoS** → *Vote for delegates to validate.*
- **PoA** → *Trust approved authority nodes.*
- **PoH** → *Prove that time has passed before validating transactions.*

6. Practical Byzantine Fault Tolerance (PBFT)

1. OverviewDefinition

Practical Byzantine Fault Tolerance (PBFT) is a **voting-based consensus algorithm** designed for **permissioned (private or consortium) blockchain networks**. It enables a distributed system to continue operating correctly even if some nodes behave maliciously, fail, or send incorrect information.

Unlike Proof of Work or Proof of Stake, PBFT does **not** use mining or staking. Instead, it relies on **multiple rounds of communication and voting** among known validator nodes to reach agreement.

PBFT was proposed in **1999** by Miguel Castro and Barbara Liskov.

Why was PBFT created?

Distributed systems often face the **Byzantine Generals Problem**, where some participants may lie, fail, or send conflicting information.

PBFT was developed to solve this problem by allowing honest nodes to reach consensus even when some nodes are faulty or malicious.

It is widely used in:

- Enterprise blockchains
 - Banking systems
 - Supply chain networks
 - Government applications
-

Core Principle

| **"Nodes exchange votes in multiple phases until a majority agrees on one valid block."**

Instead of solving puzzles or staking coins, validators communicate and vote.

2. Core Mechanism

PBFT works through **three communication phases** after a client sends a request:

A. Primary (Leader) Node

One validator acts as the **Primary (Leader)**.

Responsibilities:

- Receives client requests.
 - Proposes the next block.
 - Coordinates consensus.
-

B. Replica Nodes

The remaining validators are called **Replica Nodes**.

Responsibilities:

- Verify the proposal.
 - Exchange messages.
 - Vote.
 - Reach consensus.
-

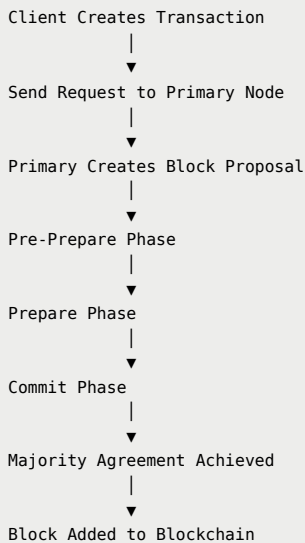
C. Three Voting Phases

PBFT uses three important phases:

1. **Pre-Prepare**
2. **Prepare**
3. **Commit**

Only after completing all three phases is the block accepted.

3. Step-by-Step Working



4. Detailed Working

Step 1 – Client Request

A user submits a transaction.

Example:

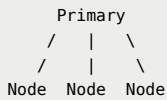
Alice → Bob
₹100

The request is sent to the **Primary Node**.

Step 2 – Pre-Prepare Phase

The Primary Node

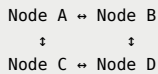
- verifies the request,
- creates a proposed block,
- broadcasts it to all replica nodes.



Step 3 – Prepare Phase

Every replica node

- checks the proposal,
- verifies the transaction,
- broadcasts a **Prepare** message to every other validator.



Every validator communicates with every other validator.

Step 4 – Commit Phase

If enough **Prepare** messages are received,
validators send **Commit** messages.

When a validator receives sufficient Commit messages,
the block is finalized.

Step 5 – Reply

Every validator informs the client that
the transaction has been committed.
Consensus is complete.

5. Participants

PBFT consists of four participants.

Client

Creates transactions.

Primary Node (Leader)

Coordinates consensus.
Proposes new blocks.

Replica Validators

Verify transactions.

Exchange votes.

Commit blocks.

Full Nodes

Maintain blockchain history.

Audit the network.

6. Selection Process

Unlike PoW,

there is

✗ No mining.

Unlike PoS,

there is

✗ No staking.

Validators are

pre-approved organizations.

The Primary Node is usually rotated periodically to avoid centralization.

7. Reward Mechanism

Most PBFT systems

do **not** issue cryptocurrency rewards.

Validators participate because

- they belong to the consortium,
- they represent organizations,
- or they provide trusted infrastructure.

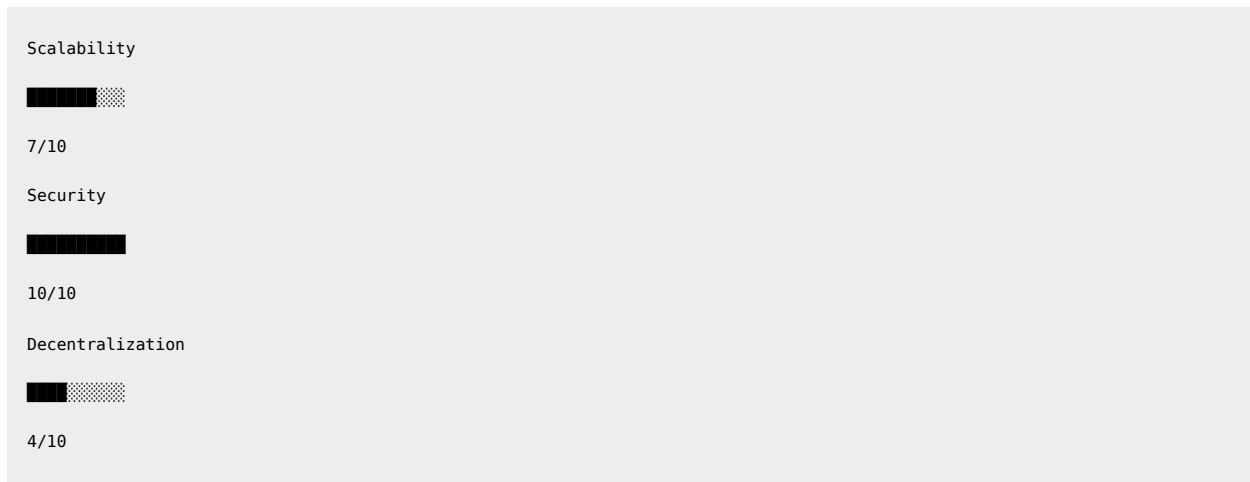
Some implementations may distribute service fees, but this is application-specific.

8. Blockchain Trilemma Analysis

Property	Score	Reason
Scalability	7/10	Efficient for small validator sets, but communication grows rapidly as the number of validators increases.
Security	10/10	Can tolerate Byzantine (malicious or faulty) nodes up to its design limit.

Property	Score	Reason
Decentralization	4/10	Permissioned validator set limits open participation.

Visual Scorecard



9. Performance

Parameter	Value
TPS	Thousands (small networks)
Block Time	Usually less than 1 second
Finality	Immediate (deterministic)
Energy Usage	Very Low
Latency	Low

Unlike PoW,
PBFT provides **immediate finality**.
Once a block is committed,
it is considered final.

10. Attack ResistanceByzantine Attack

PBFT is specifically designed to resist Byzantine failures.

It can tolerate

[
f
]

malicious nodes if

$$\begin{aligned} &[\\ n &\geq 3f + 1 \\ &] \end{aligned}$$

where:

- **n** = total validators
- **f** = maximum malicious validators

Example:

Validators	Malicious Nodes Allowed
4	1
7	2
10	3
13	4

51% Attack

Not applicable in the same way as PoW or PoS,
because PBFT relies on voting rather than mining power or stake.

Sybil Attack

Very resistant.

Unknown participants cannot become validators.

Double Spending

Extremely difficult.

Blocks are finalized through voting before acceptance.

11. Advantages

- Immediate transaction finality.
- Extremely secure against Byzantine faults.
- Very low energy consumption.
- High throughput.
- Ideal for enterprise environments.
- No mining required.

12. Disadvantages

- Communication overhead increases rapidly with more validators.
- Not suitable for thousands of validators.
- Requires permissioned membership.
- Lower decentralization than public blockchains.

13. Real-world Blockchain Mapping

Blockchain / Platform	Cryptocurrency	Why PBFT?
Hyperledger Fabric	None (enterprise platform)	Trusted organizations validate transactions quickly.
Zilliqa	ZIL	Uses PBFT within its sharded architecture for efficient consensus.
Tendermint Core (used by several blockchains)	Varies	Fast finality with Byzantine fault tolerance.
Quorum	Enterprise deployments	Supports PBFT-style consensus for permissioned networks.

14. Smart Contract Languages

Blockchain / Platform	Smart Contract Language
Hyperledger Fabric	Go, Java, JavaScript (Chaincode)
Zilliqa	Scilla
Quorum	Solidity
Tendermint-based chains	Depends on the blockchain (e.g., CosmWasm with Rust on some ecosystems)

15. Layer Classification

Blockchain	Layer
Hyperledger Fabric	Layer 1
Zilliqa	Layer 1
Quorum	Layer 1

PBFT is mainly used in **Layer 1 permissioned blockchains**.

16. Environmental Impact

Metric	Rating
Electricity	★
Carbon Footprint	Very Low
Sustainability	Excellent

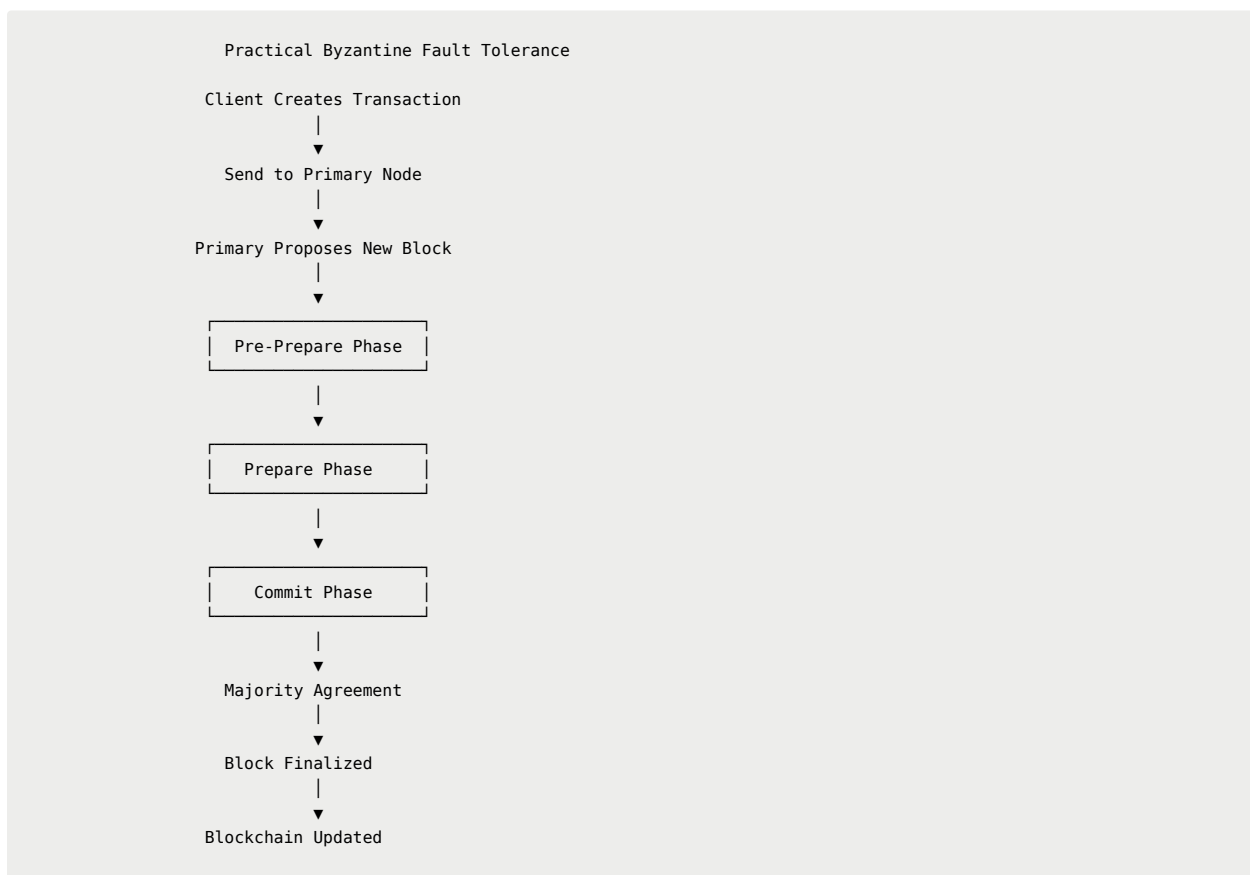
PBFT is highly energy efficient because it relies on message exchange instead of computational competition.

17. Best Use Cases

PBFT is ideal for:

- Banking systems
 - Healthcare
 - Government records
 - Supply chain management
 - Digital identity systems
 - Enterprise blockchain applications
-

18. Complete Flow Diagram



19. Summary Table

Property	Value
Consensus	Practical Byzantine Fault Tolerance
Mining	No
Staking	No
Validators	Pre-approved validator nodes
Consensus Method	Multi-phase voting
Finality	Immediate (Deterministic)
Energy Usage	Very Low
TPS	Thousands (small networks)
Block Time	Less than 1 second
Security	Excellent
Decentralization	Moderate to Low
Layer	Layer 1
Main Applications	Enterprise & Consortium Blockchains

20. Key Points for Comparison

Feature	PoW	PoS	DPoS	PoA	PoH	PBFT
Uses Mining	✓	✗	✗	✗	✗	✗
Uses Staking	✗	✓	✓	✗	✓ (with PoS)	✗
Uses Voting	✗	✗	✓	Limited (trusted validators)	✗	✓
Uses Trusted Validators	✗	✗	Elected	✓	✗	✓
Uses Cryptographic Clock	✗	✗	✗	✗	✓	✗
Finality	Probabilistic	Depends on protocol	Fast	Fast	Fast	Immediate
Energy Usage	Very High	Very Low	Very Low	Very Low	Very Low	Very Low
Speed	Low	High	Very High	Very High	Extremely High	Very High
Example Platform	Bitcoin	Ethereum	EOS	VeChain	Solana	Hyperledger Fabric

Simple Way to Remember

- **PoW** → Prove your **computing power**.
- **PoS** → Prove your **stake**.
- **DPoS** → **Vote** for delegates to validate.
- **PoA** → Trust **approved authority** validators.
- **PoH** → Use a **cryptographic clock** to order transactions before validation.

- **PBFT** → Validators **communicate and vote** in three phases (Pre-Prepare → Prepare → Commit) to tolerate malicious nodes and achieve immediate consensus.

7. Proof of Space (PoSpace) / Proof of Capacity (PoC)

1. OverviewDefinition

Proof of Space (PoSpace), also called **Proof of Capacity (PoC)**, is a consensus algorithm where participants use their **unused hard disk storage** instead of computational power to compete for the right to create the next block.

Instead of continuously solving mathematical puzzles like PoW, miners first **precompute and store cryptographic data** on their storage devices. During mining, they search this stored data to find the best solution.

Proof of Space and **Proof of Capacity** are closely related. The terms are often used interchangeably, although different blockchain implementations may use slightly different techniques.

Why was PoSpace/PoC created?

Proof of Work has two major drawbacks:

- Very high electricity consumption.
- Expensive specialized mining hardware (ASICs).

PoSpace was designed to:

- Reduce electricity usage.
 - Utilize unused storage space.
 - Lower hardware costs.
 - Make mining more environmentally friendly.
-

Core Principle

"The more storage space you dedicate to the blockchain, the greater your chance of producing the next block."

Instead of proving computational work, miners prove that they have allocated disk space.

2. Core Mechanism

PoSpace has two major phases:

A. Plotting

Before mining begins, the miner generates large amounts of cryptographic data called **plots**.

These plots are stored permanently on a hard drive.

Example:

Hard Drive

```
|
|— Plot 1
|— Plot 2
|— Plot 3
|— Plot 4
|— Plot 5
```

Plotting is done only once unless additional storage is added.

B. Farming (Mining)

When a new block is needed,

the miner searches the stored plots.

The miner with the best matching plot wins.

Unlike PoW,

no repeated hash calculations are required.

3. Step-by-Step Working

```
Generate Plots
  |
  ▼
Store on Hard Drive
  |
  ▼
New Block Challenge
  |
  ▼
Search Stored Plots
  |
  ▼
Best Match Found
  |
  ▼
Create New Block
  |
  ▼
Network Verification
  |
  ▼
Blockchain Updated
```

4. ParticipantsUsers

Create transactions.

Farmers (Miners)

Store plots.

Search plots.

Produce new blocks.

Validators

Verify blocks.

Full Nodes

Maintain the blockchain.

5. Selection Process

Selection depends on

the amount of storage dedicated to the network.

Example

Farmer A

500 GB

↓

Lower probability

Farmer B

10 TB

↓

Higher probability

More storage generally means a higher chance of creating the next block.

6. Reward Mechanism

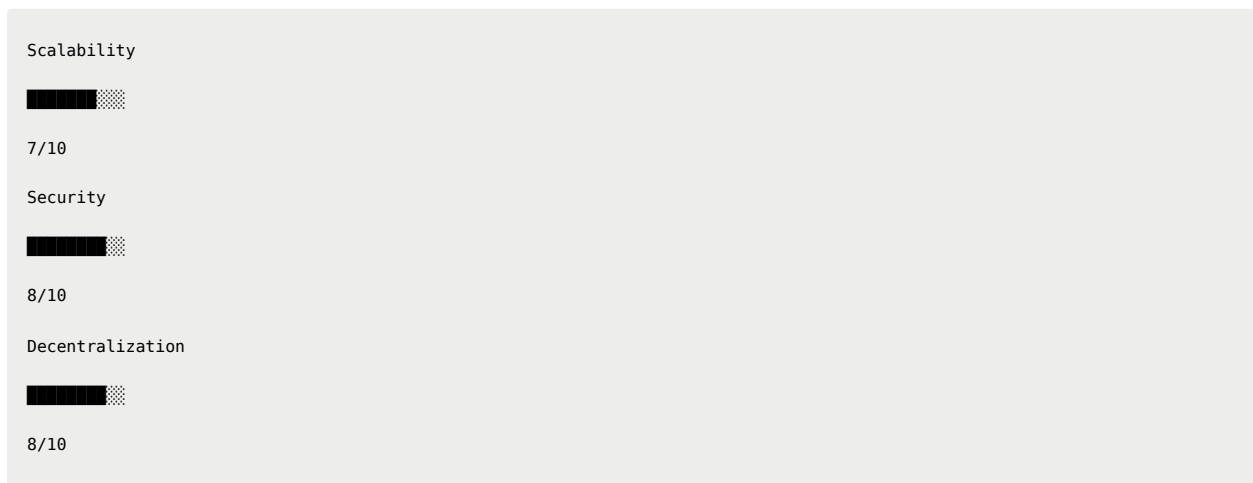
Farmers receive:

- Block rewards.
 - Transaction fees.
-

7. Blockchain Trilemma Analysis

Property	Score	Reason
Scalability	7/10	Faster and more energy-efficient than PoW, but not as fast as modern PoS systems.
Security	8/10	Large storage commitment makes attacks costly.
Decentralization	8/10	Ordinary users can participate using consumer storage devices.

Visual Scorecard



8. Performance

Parameter	Value
TPS	Moderate (depends on implementation)
Block Time	A few minutes (varies by blockchain)
Finality	Moderate
Energy Usage	Low
Hardware	Hard disks or SSDs

9. Attack Resistance

51% Attack

An attacker would need to control most of the storage participating in the network.

This is expensive but generally less energy-intensive than acquiring majority hash power in PoW.

Sybil Attack

Creating fake identities alone provides no advantage.

Actual storage capacity is required.

Double Spending

Very difficult after confirmation.

10. Advantages

- Very low electricity consumption.
 - Uses unused disk storage.
 - Lower hardware costs than ASIC mining.
 - Environmentally friendly.
-

11. Disadvantages

- Requires significant storage capacity.
 - Initial plotting can be time-consuming.
 - Large-scale storage farms may gain advantages.
-

12. Real-world Blockchain Mapping

Blockchain	Cryptocurrency	Why PoSpace?
Chia	XCH	Designed to replace energy-intensive mining with storage-based farming.
Burst	BURST	Early blockchain using Proof of Capacity.

13. Smart Contract Languages

Blockchain	Language
Chia	Chialisp
Burst	Limited smart contract capabilities

14. Layer Classification

Blockchain	Layer
Chia	Layer 1
Burst	Layer 1

15. Environmental Impact

Metric	Rating
Electricity	★★

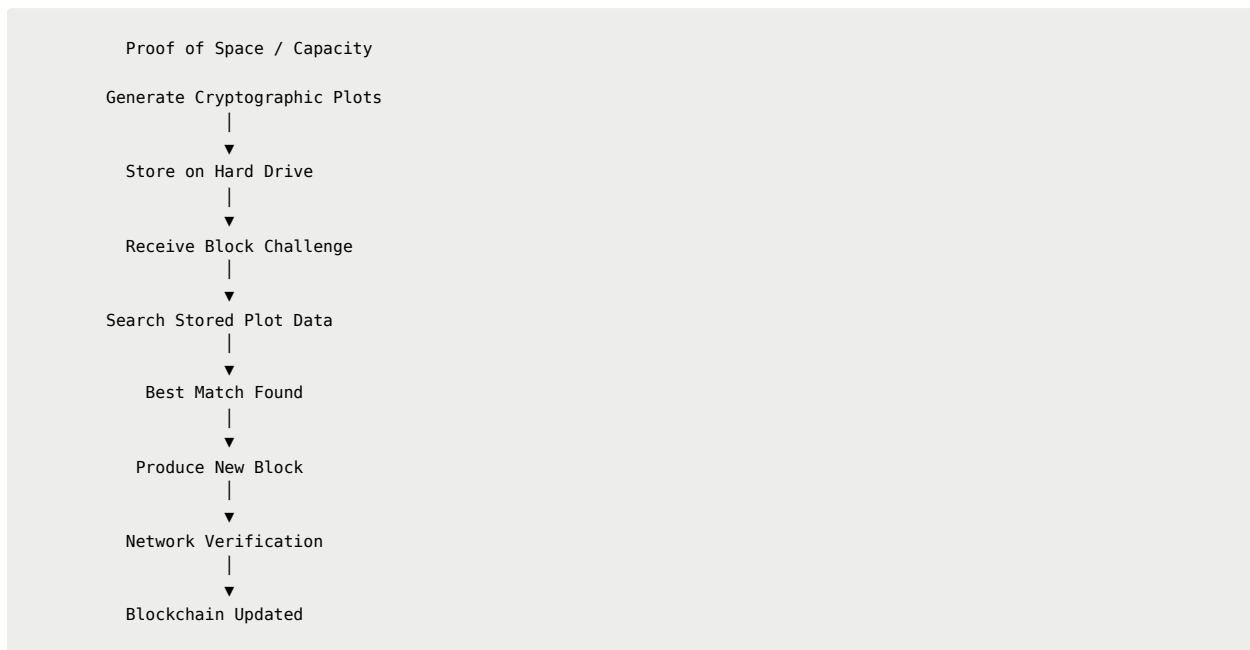
Metric	Rating
Carbon Footprint	Low
Sustainability	High

PoSpace consumes much less electricity than PoW because it mainly reads stored data instead of performing continuous computations.

16. Best Use Cases

- Green blockchain networks.
- Sustainable cryptocurrencies.
- Long-term decentralized storage ecosystems.

17. Complete Flow Diagram



18. Summary Table

Property	Value
Consensus	Proof of Space / Capacity
Mining	Storage-based farming
Uses Staking	No
Main Resource	Hard Disk Storage
Energy	Low

Property	Value
Speed	Moderate
Security	High
Layer	Layer 1
Example	Chia

8. Proof of Burn (PoB)

1. OverviewDefinition

Proof of Burn (PoB) is a consensus algorithm where participants **permanently destroy (burn)** a portion of their cryptocurrency to earn the right to validate transactions and create new blocks.

Burning coins demonstrates a long-term commitment to the network because the coins can never be recovered.

Why was PoB created?

PoB was introduced to:

- Eliminate energy-intensive mining.
- Replace hardware investment with economic commitment.
- Reduce inflation by decreasing the total coin supply.
- Encourage long-term participation.

Core Principle

|"The more coins you permanently burn, the greater your opportunity to become a block producer."

Burning coins is like making a non-refundable investment in the network.

2. Core Mechanism

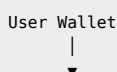
PoB works in three stages.

A. Burn Coins

Validators send coins to a **burn address**.

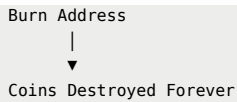
A burn address has **no private key**, making the coins permanently inaccessible.

Example:



```

graph TD
    UW[User Wallet] --> BA[ ]
    style BA fill:none,stroke:none
  
```



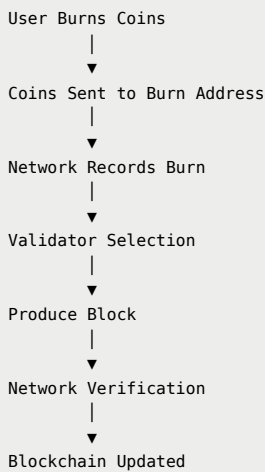
B. Gain Mining Rights

After burning coins,
the participant earns the ability to create future blocks.
Generally, burning more coins increases the chance of selection.

C. Validate Blocks

Selected participants validate transactions and produce blocks.

3. Step-by-Step Working



4. ParticipantsUsers

Create transactions.

Burners (Validators)

Burn coins.
Validate blocks.
Receive rewards.

Full Nodes

Verify transactions.

Maintain the blockchain.

5. Selection Process

Selection depends on
the amount of cryptocurrency burned.

Example

```
Validator A  
  
Burns  
  
50 Coins  
  
↓  
  
Lower probability
```

```
Validator B  
  
Burns  
  
500 Coins  
  
↓  
  
Higher probability
```

6. Reward Mechanism

Validators receive:

- Block rewards.
- Transaction fees.

Although they destroy coins initially,
they can recover value over time through rewards if they continue participating successfully.

7. Blockchain Trilemma Analysis

Property	Score	Reason
Scalability	7/10	No mining competition improves efficiency over PoW.
Security	8/10	Burning coins creates an economic cost for attackers.
Decentralization	7/10	Participation is open, but wealthy users may burn more coins.

Visual Scorecard

Scalability



7/10

Security



8/10

Decentralization



7/10

8. Performance

Parameter	Value
TPS	Moderate
Block Time	Depends on implementation
Finality	Moderate
Energy Usage	Very Low

9. Attack Resistance^{51% Attack}

Attackers would need to destroy a significant amount of cryptocurrency to gain influence.

Sybil Attack

Fake identities provide no advantage.

Actual economic sacrifice is required.

Double Spending

Difficult once blocks are confirmed.

10. Advantages

- Very low energy consumption.
- Encourages long-term commitment.
- Reduces circulating coin supply.
- No expensive mining hardware.

11. Disadvantages

- Burning coins is irreversible.
- Wealthier participants can burn more coins.
- Less widely adopted than PoW or PoS.

12. Real-world Blockchain Mapping

Blockchain	Cryptocurrency	Why PoB?
Slimcoin	SLM	Designed to use Proof of Burn as its primary consensus mechanism.
Counterparty	XCP	Used Bitcoin burning during its initial token distribution (not as ongoing consensus).

Note: Pure PoB consensus is relatively rare in modern public blockchains.

13. Smart Contract Languages

Blockchain	Language
Slimcoin	No major smart contract platform
Counterparty	Bitcoin Script (limited programmability)

14. Layer Classification

Blockchain	Layer
Slimcoin	Layer 1
Counterparty	Built on Bitcoin (Layer built over Bitcoin rather than an independent Layer 1)

15. Environmental Impact

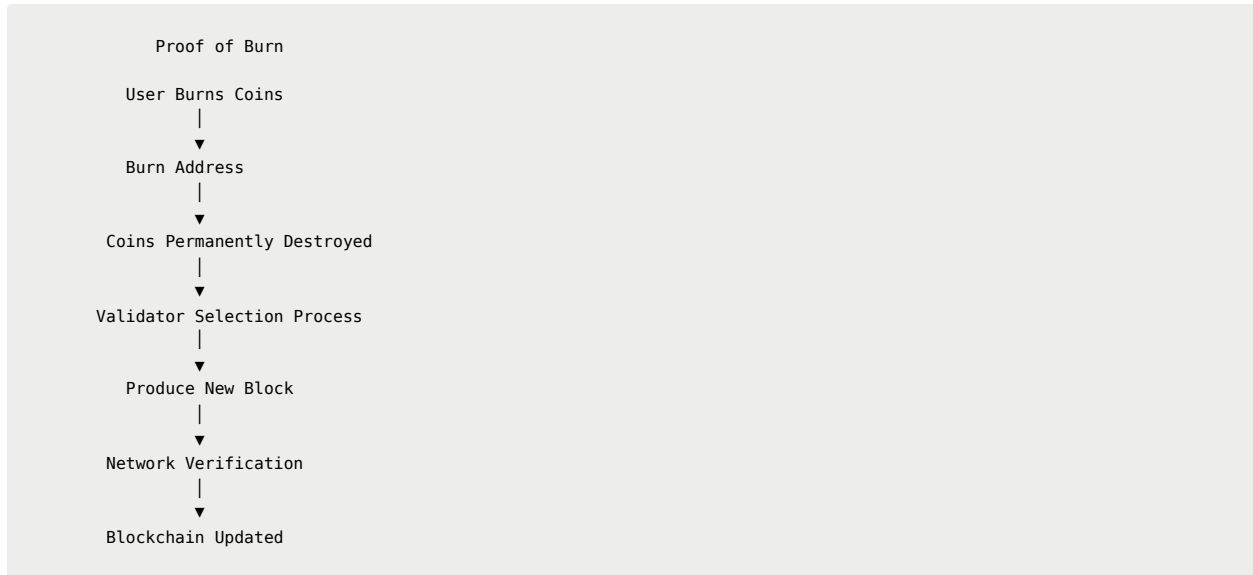
Metric	Rating
Electricity	★
Carbon Footprint	Very Low
Sustainability	Excellent

16. Best Use Cases

- Experimental blockchain consensus.
- Deflationary cryptocurrency models.

- Networks emphasizing long-term commitment.

17. Complete Flow Diagram



18. Summary Table

Property	Value
Consensus	Proof of Burn
Mining	No
Uses Staking	No
Main Resource	Burned Cryptocurrency
Energy	Very Low
Speed	Moderate
Security	High
Layer	Layer 1
Example	Slimcoin

Easy Way to Remember

- **PoSpace / PoC → Prove your storage capacity.**
- **PoB → Prove your commitment by permanently burning coins.**

9. Proof of Elapsed Time (PoET)

1. Overview

Proof of Elapsed Time (PoET) is a consensus algorithm in which each validator waits for a **randomly assigned amount of time**, and the validator whose timer expires first earns the right to create the next block.

Unlike Proof of Work, validators do not compete by solving computational puzzles. Instead, every validator receives a random waiting time generated inside a **Trusted Execution Environment (TEE)**.

PoET was developed by **Intel** for enterprise blockchain applications.

Why was PoET created?

PoW consumes large amounts of electricity.

PoS requires staking cryptocurrency.

Enterprise organizations wanted a consensus algorithm that:

- Uses minimal computing power.
- Is fair.
- Is fast.
- Works in permissioned networks.

PoET satisfies these requirements.

Core Principle

"Every validator receives a random waiting time. The validator whose timer finishes first creates the next block."

2. Core Mechanism

PoET depends on a **Trusted Execution Environment (TEE)** such as **Intel SGX (Software Guard Extensions)**.

A. Trusted Execution Environment

The TEE securely generates a random waiting time.

Validators cannot manipulate or predict the timer.

B. Random Waiting Time

Example:

Validator A → 12 seconds

Validator B → 5 seconds

Validator C → 18 seconds

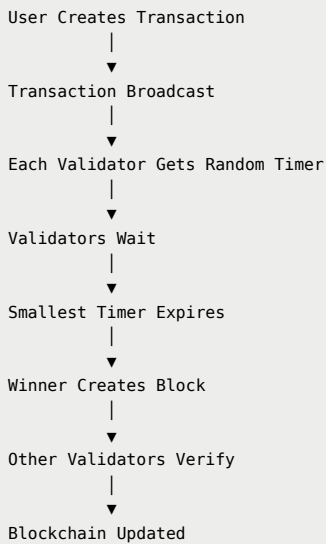
Validator B wins because its timer expires first.

C. Block Creation

The winning validator creates the next block.

Other validators verify the proof generated by the TEE.

3. Step-by-Step Working



4. ParticipantsUsers

Create transactions.

Validators

Receive waiting times.

Wait for timer expiration.

Create blocks.

Trusted Execution Environment

Generates secure random waiting times.

Full Nodes

Maintain blockchain.

Verify blocks.

5. Selection Process

Validators are selected based on
random waiting time.

Example

Validator A

15 sec

Validator B

4 sec

Winner

Validator C

10 sec

6. Reward Mechanism

Validators receive:

- Block rewards (if implemented).
- Transaction fees.

Enterprise blockchains often use PoET without cryptocurrency rewards.

7. Blockchain Trilemma Analysis

Property	Score	Reason
Scalability	9/10	Very fast because there is no mining competition.
Security	8/10	Secure if the Trusted Execution Environment is trusted.
Decentralization	5/10	Requires trusted hardware and permissioned participation.

Visual Scorecard

Scalability



9/10

Security



8/10

Decentralization



5/10

8. Performance

Parameter	Value
TPS	High
Block Time	A few seconds
Finality	Fast
Energy Usage	Very Low
Hardware	Trusted Execution Environment

9. Attack Resistance51% Attack

Less relevant because validators are pre-approved.

Sybil Attack

Very resistant.

Unauthorized users cannot become validators.

Hardware Attack

If the Trusted Execution Environment is compromised, security could be affected.

This is the primary concern with PoET.

10. Advantages

- Extremely low energy usage.
- Fast block creation.
- Fair validator selection.
- Suitable for enterprise blockchains.

11. Disadvantages

- Requires trusted hardware.
- Depends on Intel SGX or similar technology.
- Less decentralized than public blockchains.

12. Real-world Blockchain Mapping

Blockchain	Cryptocurrency	Why PoET?
Hyperledger Sawtooth	None	Uses PoET to provide efficient enterprise consensus.

13. Smart Contract Languages

Blockchain	Language
Hyperledger Sawtooth	Python
Hyperledger Sawtooth	Go
Hyperledger Sawtooth	JavaScript
Hyperledger Sawtooth	Rust

14. Layer Classification

Blockchain	Layer
Hyperledger Sawtooth	Layer 1

15. Environmental Impact

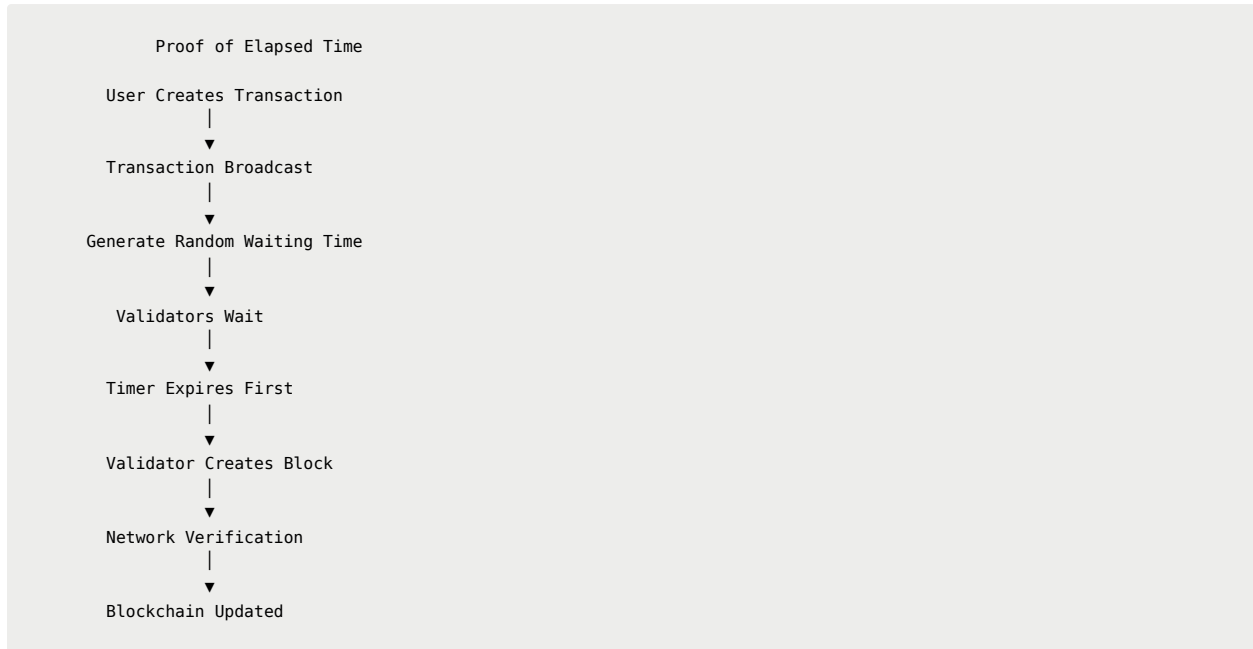
Metric	Rating
Electricity	★
Carbon Footprint	Very Low
Sustainability	Excellent

16. Best Use Cases

- Enterprise blockchain.
- Supply chain.
- Healthcare.
- Manufacturing.

- Banking.

17. Complete Flow Diagram



18. Summary Table

Property	Value
Consensus	Proof of Elapsed Time
Mining	No
Staking	No
Main Resource	Trusted Hardware
Energy	Very Low
Speed	High
Security	High
Layer	Layer 1
Example	Hyperledger Sawtooth

10. Proof of Importance (PoI)

1. OverviewDefinition

Proof of Importance (Pol) is a consensus algorithm that selects validators based not only on the amount of cryptocurrency they hold but also on **how actively they participate in the blockchain network**.

Unlike Proof of Stake, where wealth is the primary factor, Pol rewards users who actively contribute to the network.

It was introduced by the NEM blockchain.

Why was Pol created?

PoS tends to favor wealthy users.

Pol was designed to encourage:

- Active transactions.
 - Network participation.
 - Fair validator selection.
 - Better decentralization.
-

Core Principle

| "Validators are selected based on an importance score rather than only their coin balance."

2. Core Mechanism

Pol calculates an **Importance Score**.

The score considers:

A. Amount of Coins Held

More coins increase importance.

B. Transaction Activity

Users who regularly transact receive higher scores.

C. Network Contribution

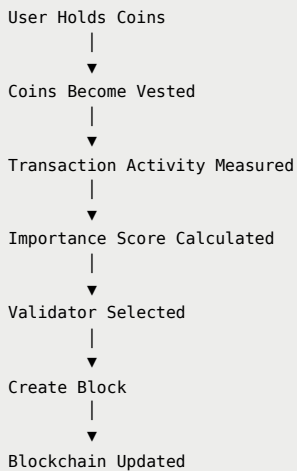
Sending transactions to many participants improves importance.

D. Vested Balance

Only coins held for a certain period are counted.

This prevents users from borrowing coins briefly to gain influence.

3. Step-by-Step Working



4. Participants

Create transactions.

Active Users

Increase their importance score through participation.

Harvesters (Validators)

Create new blocks.

NEM uses the term **Harvesting** instead of Mining.

Full Nodes

Verify blockchain.

5. Selection Process

Selection depends on

Importance Score.

Example

User A
1000 Coins
Few Transactions
↓
Importance

40

User B

700 Coins

Many Transactions

↓

Importance

70

Winner

User B may be selected despite holding fewer coins.

6. Reward Mechanism

Harvesters receive:

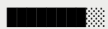
- Transaction fees.
- Block rewards (depending on network design).

7. Blockchain Trilemma Analysis

Property	Score	Reason
Scalability	8/10	Efficient validator selection without mining.
Security	8/10	Combines stake with activity, making manipulation harder.
Decentralization	9/10	Encourages broader participation instead of rewarding only large holders.

Visual Scorecard

Scalability



8/10

Security



8/10

Decentralization



9/10

8. Performance

Parameter	Value
TPS	Moderate to High
Block Time	About 1 minute (NEM)
Finality	Fast
Energy Usage	Very Low

9. Attack Resistance51% Attack

More difficult because influence depends on stake **and** network participation.

Sybil Attack

Creating fake identities provides little benefit.

Meaningful transaction history and vested balance are required.

Double Spending

Protected through validator consensus and transaction confirmation.

10. Advantages

- Rewards active users.
- Encourages healthy network activity.
- Energy efficient.
- Better decentralization than simple PoS.

11. Disadvantages

- Importance score calculation is more complex.
- Less widely adopted than PoW or PoS.
- Mainly associated with the NEM ecosystem.

12. Real-world Blockchain Mapping

Blockchain	Cryptocurrency	Why Pol?
NEM	XEM	Designed to reward active participation instead of only wealth.

13. Smart Contract Languages

Blockchain	Language
NEM	Smart Asset System (built-in blockchain features rather than a traditional general-purpose smart contract language)

14. Layer Classification

Blockchain	Layer
NEM	Layer 1

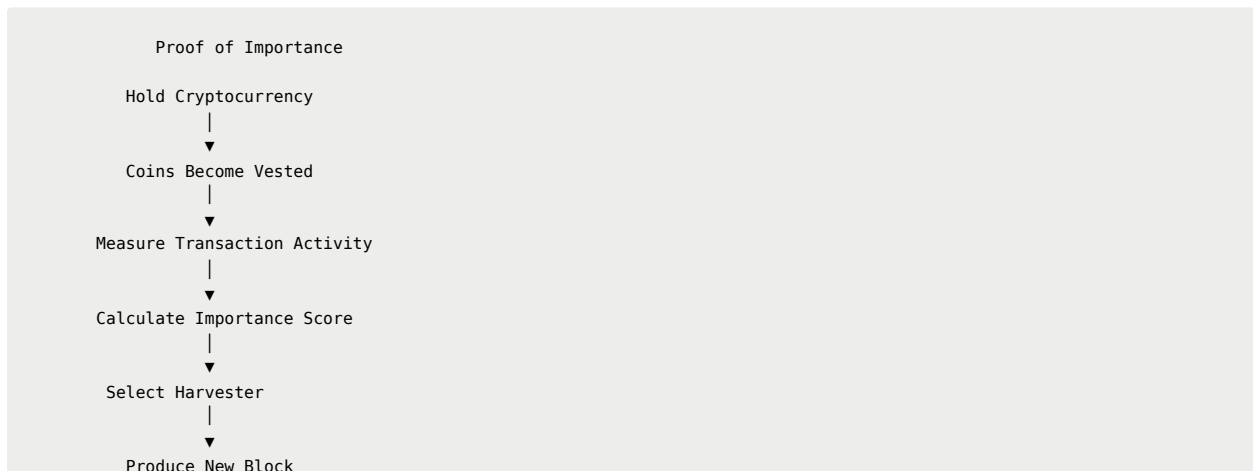
15. Environmental Impact

Metric	Rating
Electricity	★
Carbon Footprint	Very Low
Sustainability	Excellent

16. Best Use Cases

- Community-driven blockchain networks.
- Payment systems.
- Asset management.
- Business applications.

17. Complete Flow Diagram



↓
Blockchain Updated

18. Summary Table

Property	Value
Consensus	Proof of Importance
Mining	No
Staking	Yes (with activity-based scoring)
Main Resource	Importance Score
Energy	Very Low
Speed	High
Security	High
Layer	Layer 1
Example	NEM

Final Comparison of All 10 Consensus Algorithms

Algorithm	Main Resource	Block Creator	Energy	Speed	Best Example
PoW	Computing Power	Miner	● Very High	Slow	Bitcoin
PoS	Staked Coins	Validator	● Very Low	High	Ethereum
DPoS	Voting + Stake	Delegate	● Very Low	Very High	EOS
PoA	Trusted Identity	Authority Node	● Very Low	Very High	VeChain
PoH	Cryptographic Time + PoS	Leader Validator	● Very Low	Extremely High	Solana
PBFT	Multi-phase Voting	Validators	● Very Low	Very High	Hyperledger Fabric
PoSpace / PoC	Storage Space	Farmer	● Low	Moderate	Chia
PoB	Burned Coins	Burner / Validator	● Very Low	Moderate	Slimcoin
PoET	Random Waiting Time (TEE)	Validator	● Very Low	High	Hyperledger Sawtooth
Pol	Importance Score	Harvester	● Very Low	High	NEM